ISOMETRIC

Bio-oil Geological Storage

This protocol outlines the MRV and best practices for high-quality carbon removal via the conversion of biomass to bio-oil and injection into natural or engineered subsurface features and geologic formations.

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1.0 Introduction

This Protocol provides the requirements and procedures for the calculation of net CO_2e removal from the atmosphere via the conversion of biomass to bio-oil (and co-products) and injection of bio-oil into natural or engineered subsurface features or geologic formations which may include, but are not limited to, reservoirs, saline aquifers, caverns or mines, for long term sequestration of atmospheric CO_2. Bio-oil Carbon Capture & Sequestration is considered a subsector of Biomass Carbon Removal and Storage (BiCRS). This Protocol applies to bio-oil sequestration technologies or projects, which typically consist of activities associated with three sub-processes - biomass growth, biomass conversion, and bio-oil injection and storage.

Bio-oil is a mixture of water, organic acids, aldehydes, ketones, sugars, phenols, and other organic compounds derived from the thermal breakdown of biomass[1]. Thermal breakdown of biomass is achieved via thermochemical processes, such as pyrolysis, liquefaction, or gasification, which heat biomass in low- or no- oxygen environments to high temperature (approximately 350°-650°C). Bio-oil is often also referred to as pyrolysis oil or bio-crude.

Bio-oil can have co-products like biochar mixed into it ahead of injection underground. Within this Protocol, we use the words ‘bio-oil’, ‘bio-oil with biochar’ and ‘injectant’ interchangeably.

The Protocol accounts for quantification of the gross amount of CO_2 removed via injection of bio-oil into geologic formations, as well as the accounting for all greenhouse gas (GHG) emissions associated with the growth and collection of biomass feedstock, biomass conversion, biomass injection process, all transportation and embodied emissions associated with the process, and emissions associated with leakage. The GHG Statement is considered a cradle-to-grave analysis.

This Protocol is developed to adhere to the requirements of ISO 14064-2: 2019 – Greenhouse Gases – Part 2: Specification with guidance at the project level for quantification, monitoring, and reporting of greenhouse gas emission reductions or removal enhancements. The Protocol ensures:

- consistent, accurate procedures are used to measure and monitor all aspects of the process required to enable accurate accounting of net CO_2e removals
- consistent system boundaries and calculations are utilized to quantify net CO_2e removal for bio-oil injection projects
- requirements are met to ensure the CO_2 removals are additional
- evidence is provided and verified by independent third parties to support all net CO_2e removal claims

2.0 Sources and Reference Standards & Methodologies

Specific Standards and protocols which are utilized as the foundation of this Protocol and for which this Protocol is intended to be fully compliant with are the following:

- Isometric Standard
- ISO 14064-2: 2019 – Greenhouse Gases – Part 2: Specification with guidance at the project level for quantification, monitoring, and reporting of greenhouse gas emission reductions or removal enhancements

Additional reference standards that inform the requirements and overall practices incorporated in this Protocol include:

- ISO 14064-3: 2019 – Greenhouse Gases – Part 3: Specification with Guidance for the verification and validation of greenhouse gas statements
- ISO 14040: 2006 - Environmental Management - life cycle Assessment - Principles & Framework
- ISO 14044: 2006 - Environmental Management - life cycle Assessment - Requirements & Guidelines

Additional standards, methodologies, and protocols that were reviewed, referenced, or for which attempts to align with or leverage in development of this Protocol include:

- Bio-oil Sequestration Prototype Protocol for Measurement, Reporting, & Verification. Carbon Direct / EcoEngineers, 2022
- Carbon Capture & Sequestration Protocol under the Low Carbon Fuel Standard. California Air Resources Board (CARB). August, 2018

3.0 Future Versions

This Protocol was developed based on the current state of the art and current publicly available science regarding biomass conversion and bio-oil injection. Because bio-oil injection and storage in geologic formations is a novel carbon dioxide removal (CDR) approach, with limited published literature, the Protocol incorporates requirements that may be more stringent than some current relevant regulations for underground injection, or other protocols related to biomass utilization for CDR.

This approach, notably when specifying requirements for demonstrating bio-oil storage durability or permanence, will likely be altered in future versions of the Protocol as the stability of bio-oil in geologic formations becomes well demonstrated and documented, reversal risks are proven to be limited or non-existent, and the overall body of knowledge and data regarding all processes, from feedstock supply, to conversion, and to permanent storage is significantly increased.

4.0 Applicability

This Protocol applies to projects or processes which:

- utilize agricultural or forestry residues as eligible feedstocks in accordance with the framework set out in the Biomass Feedstock Accounting Module v1.2
- convert the biomass to bio-oil via pyrolysis or similar processes or utilize bio-oil produced by a third-party supplier
- inject the bio-oil into natural or engineered geologic formations for long duration storage purposes via an underground injection well

This Protocol applies to projects and associated operations that meet all of the following project conditions:

- the project provides a net-negative CO_2e impact (net CO_2e removal) as calculated in the GHG Statement, in compliance with Section 7
- the biomass feedstock utilized is sustainably sourced[2]
- the project does no net harm to the environment and society
- the project is considered additional, in accordance with the requirements of Section 6.4
- the project provides long duration storage (>1000 yr estimated) of carbon in geologic formations
- the geologic storage site is located in the US
- the geologic storage site is properly permitted and has a current relevant UIC well permit [3]. The site must be operated in compliance with current permits including those issued by the US EPA or U.S. States for underground injection control wells and specifically identify bio-oil or an equivalent type of injectant, as acceptable injectants under the permit [4].

Projects that are explicitly NOT eligible include the following:

- Projects that utilize bio-oil as part of enhanced hydrocarbon recovery (EHR or EHR+)
- Projects that utilize biomass that doesn't meet the eligibility criteria detailed in Biomass Feedstock Accounting Module v1.2.

5.0 Environmental and Social Safeguarding

The project must consider other environmental and social impacts and the Project Proponent must provide evidence that the project will do no net environmental or socioeconomic harm, complying with Section 3.7 of the Isometric Standard.

In addition to that outlined with the Isometric Standard, it must consider all aspects of the project from feedstock growth through injection and storage. At a minimum the Project Proponent must:

1. document activities conducted under the project which would require it to obtain environmental permits and provide a listing of all permits or construction approvals received or applied for and their current status and compliance history. This may include, but is not limited to, activities under the Resource Conservation & Recovery Act (RCRA), the U.S. EPA Underground Injection Control (UIC) program, the National Pollutant Discharge Elimination System (NPDES) program under the Clean Water Act (CWA), the Prevention of Significant Deterioration (PSD) program under the Clean Air Act (CAA), and other relevant environmental permits such as federal, state, county, or city permits for air, water or waste discharges;
2. document activities that would require the Project Proponent to obtain any drilling permits, access agreements, or any encroachment permits under county or city guidelines, or any federal, state, or local air, water, or restricted land use operating permits;
3. when agricultural residues are used, provide documentation of characterization of biomass in accordance with the Biomass Feedstock Accounting Module v1.2 to demonstrate impact on carbon and nutrient (nitrogen, phosphorus, potassium) removal;
4. provide documentation demonstrating whether the bio-oil is non-hazardous or hazardous, based on U.S. EPA RCRA requirements and testing completed by an accredited laboratory in accordance with U.S. EPA methods recommended in the Test Methods for Evaluating Solid Waste: Physical/Chemical Methods Compendium (SW-846)[5] or equivalent, unless exempted by permit rule or standard. If the bio-oil is classified as hazardous:

- evidence must be provided that all protocols for hazardous substance handling and management are being followed.
- injection well permits must allow for hazardous waste injection and evidence that the permit conditions are being followed, must be provided.
- containment within the targeted storage reservoir must be ensured and monitored, and all evidence provided;

5. demonstrate a risk assessment and mitigation plan for safeguarding working conditions, especially when it comes to safe handling and transportation of biomass feedstocks as well as workers operating the storage facility.

6.0 Relation to the Isometric Standard

The following topics are covered briefly in this Protocol due to their inclusion in the Isometric Standard, which governs all Isometric Protocols. See in-text references to the Isometric Standard for further guidance.

6.1 Project Design Document

For each specific project to be evaluated under this Protocol, the Project Proponent must document project characteristics in a Project Design Document (PDD) as outlined in Section 3.2 of the Isometric Standard. The PDD will form the basis for project verification and evaluation in accordance with this Protocol, and must include consideration of processes unique to bio-oil such as:

- location information for biomass production, biomass conversion, bio-oil injection, and geologic storage formation
- conditions of biomass use prior to project initiation
- details on technologies, products, and services relevant to biomass conversion processes, including production rates and volumes

6.2 Validation and Verification

Projects must be validated and project net CO_2e removals verified by an independent third party consistent with the requirements described in this Protocol as well as in Section 4 of the Isometric Standard.

The Validation and Verification Body (VVB) must consider following requisite components:

1. Validate that feedstock adheres to the requirements listed in the Biomass Feedstock Accounting Module v1.2.
2. Verify that storage sites adhere to the requirements listed in the relevant storage module.

3. Verify that the quantification approach and monitoring plan adheres to requirements of Section 7, including demonstration of required records.
4. Verify that the Environmental & Social Safeguards outlined in Section 5 are met.
5. Verify that the project is compliant with requirements outlined in the Isometric Standard.

6.2.1 Verification Materiality

The threshold for Materiality, considering the totality of all omissions, errors and mis-statements, is 5%, in accordance with Section 4.3 of the Isometric Standard.

Verifiers should also verify the documentation of uncertainty of the GHG Statement as required by Section 2.5.7 of the Isometric Standard. Qualitative Materiality issues may also be identified and documented, such as[5]:

- control issues that erode the verifier's confidence in the reported data;
- poorly managed documented information;
- difficulty in locating requested information;
- noncompliance with regulations indirectly related to GHG emissions, removals or storage

6.2.2 Site Visits

Project validation and verification must incorporate site visits to project facilities in accordance with the requirements of ISO 14064-3, 6.1.4.2, including, at a minimum, site visits during validation and initial verification to the biomass conversion site and the bio-oil injection site. Validators should, whenever possible, observe operation of the conversion and injection processes to ensure full documentation of process inputs and outputs through visual observation.

A site visit must occur at least once every 2 years at each location.

6.2.3 Verifier Qualifications & Requirements

Verifiers and validators must comply with the requirements defined in Section 4 of the Isometric Standard. In addition, teams shall maintain and demonstrate expertise associated with the specific technologies of interest, including biomass growth, biomass conversion, bio-oil production and geologic storage.

Competency must be demonstrated through the relevant sectoral scope accreditations listed below, based on IAF MD 14 and in accordance with Isometric's VVB policy:

- Storage - Carbon Capture and Storage of CO_2 in Geological Formations

6.3 Ownership

CDR via bio-oil production and injection is often a result of a multi-step process (such as biomass growth, harvesting, transport, conversion, injection, and storage), with activities in each step managed and operated by a different operator, company, or owner. When there are multiple parties involved in the process (e.g., forestry owner or injection site operator), and to avoid double counting of net CO_2e removals, a single Project Proponent must be specified as the sole owner of the Credits. Contracts must comply with all requirements defined in Section 3.1 of the Isometric Standard.

6.4 Additionality

The Project Proponent shall be able to demonstrate additionality through compliance with Section 2.5.3 of the Isometric Standard. The baseline scenario and counterfactual utilized to assess additionality must be project-specific, and are described Section 7.2 of this Protocol.

Additionality determinations should be reviewed and completed every two years, at a minimum, or whenever project operating conditions change significantly, such as the following:

- regulatory requirements or other legal obligations for project implementation change or new requirements are implemented
- project financials indicate carbon finance is no longer required, potentially due to, for example
  - increased tipping fees for waste feedstocks
  - sale of co-products that make the business viable without carbon finance
  - reduced rates for capital access

Any review and change in the determination of additionality shall not affect the availability of carbon finance and carbon credits for the current or past Crediting Periods, but, if the review indicates the project has become non-additional, this shall make the project ineligible for future credits[6].

6.5 Uncertainty

The uncertainty in the overall estimate of the net CO_2e removal as a result of the project must be accounted for. The total net CO_2e removed ($CO_2e_{Removal}$) for a specific batch (n) must be conservatively determined based on the relevant requirements outlined in Section 2.5.7 of the Isometric Standard.

6.5.1 Reporting of uncertainty

Projects must report a list of all input variables used in the net CO_2e removal calculation and their uncertainties, including:

- emission factors utilized, as published in public and other databases used
- values of measured parameters from process instrumentation, such as truck weights from weigh scales, flow rates from flow meters, electricity usage from utility power meters, and other similar equipment
- laboratory analyses, including analysis of carbon content of injected bio-oil

The uncertainty information should at least include the minimum and maximum values of each variable that goes into the net CO_2e removal calculation. More detailed uncertainty

information should be provided if available, as outlined in Section 2.5.7 of the Isometric Standard.

In addition, a sensitivity analysis that demonstrates the impact of each input parameter's uncertainty on the final net CO_2e uncertainty must be provided. Details of the sensitivity analysis method must be provided such that a third party can reproduce the results. Input variables may be omitted from an uncertainty analysis if they contribute to a < 1% change in the net CO_2e removal. For all other parameters, information about uncertainty must be specified.

6.6 Data sharing

In accordance with the Isometric Standard, all evidence and data related to the underlying quantification of net CO_2e removal will be available to the public through Isometric's platform. That includes:

- Project Design Document
- GHG Statement
- Measurements taken and supporting documentation, such as calibration certificates
- Emission factors used
- Scientific literature used

The Project Proponent can request certain information to be restricted (only available to authorized Buyers, the Registry and VVB) where it is subject to confidentiality. This includes emissions factors from licensed databases. However, all other numerical data produced or used as part of the quantification of net CO_2e removal will be made available.

7.0 Quantification of net CO_2e removal

7.1 System Boundary & GHG Emissions Scope

The scope of this Protocol includes GHG sources, sinks, and reservoirs (SSRs) associated with a bio-oil injection CDR project. A cradle-to-grave GHG Statement must be prepared encompassing the GHG emissions relating to SSRs controlled by and related to the project, including the following activities:

- Biomass production (growth and harvesting)
- Biomass transport
- Biomass conversion (bio-oil production)
- Bio-oil transport to processing sites and injection site
- Bio-oil injection
- Monitoring (including carbon content monitoring, reversals monitoring, lab processing and surveys)

It should be noted that the physical and system boundaries for the bio-oil injection and storage site should include the following subprocess:

- Injection facility (including any surface processing or preparation of bio-oil and injectants, injection system, and balance of plant or auxiliary systems)
- Monitoring wells and any above ground monitoring systems or devices
- The limits of the geologic storage reservoir (vertical and lateral)
- Embodied emissions associated with each activity and process, such as for manufacture and shipping of process equipment and for consumables used

Emissions for processes within the system boundary shall include all GHG SSRs from the construction or manufacturing of each physical site and associated equipment, closure and disposal of each site and associated equipment, and operation of each process (biomass production, biomass conversion, bio-oil injection), to include embodied emissions of equipment and consumables in the process. Any emissions required to

source the feedstock must be accounted for. These include feedstock collection/harvest, preparation, and transportation. The Project Proponent is responsible for identifying all GHG SSRs directly or indirectly related to project activities.

Any emissions from sub-processes or process changes that would not have taken place without the involvement of the CDR process, such as subsequent transportation and refining, must be fully considered in the system boundary. This allows for accurate consideration of additional, incremental emissions induced by the CDR process.

Ancillary activities (such as supplementary research and development activities and corporate administrative activities) that are associated with a project but are not directly or indirectly related to the issuance of Credits can be excluded from the system boundary.

In addition, biomass conversion pyrolysis processes will typically produce co-products of biochar and ash, and potentially other products. All emissions associated with the entire system where bio-oil and co-products are produced must be allocated to the CDR process. When biochar is mixed into bio-oil ahead of injection, emissions associated with all activities related to biochar production must be accounted for in totality.

The system boundary for GHG accounting must include all SSRs controlled by and related to the project, including but not limited to the SSRs in Table 1.

Table 1. Scope of activities and GHG SSRs to be included by the removal project

EMISSION CATEGORY	GHG SOURCE, SINK OR RESERVOIR	GHG	INCLUDED?	JUSTIFICATION
Biomass feedstock	Fuel Use (harvesting and/or collection)	CO_2	Yes	Primary emission from fuel combustion
		CH_4	Yes	Potential release during fuel combustion; included for completeness
		N_2O	Yes	

EMISSION CATEGORY	GHG SOURCE, SINK OR RESERVOIR	GHG	INCLUDED?	JUSTIFICATION
	Electricity Use (harvesting and/or collection)	CO_2	Yes	Primary emission from electricity generation
		CH_4	Yes	Potential release during electricity generation; included for completeness
		N_2O	Yes	
	Embodied emissions – equipment manufacture, construction, demolition	CO_2	Yes	Primary emission from manufacture of equipment due to energy consumption
		CH_4	Yes	CH_4 release possible depending on manufacturing process and energy consumption
		N_2O	Yes	Included for completeness. May be demonstrated as negligible and excluded
	Land Use Change	CO_2	Yes	Accounting framework outlined in the Biomass Feedstock Accounting Module v1.2
		CH_4	Yes	
		N_2O	Yes	
Biomass Processing	Electricity Use	CO_2	Yes	Primary emission from electricity generation
		CH_4	Yes	Potential release during electricity generation; included for completeness
		N_2O	Yes	
	Process Emissions	CO_2	Yes	Release possible depending on processing method, e.g. tail gases released during pyroylsis
		CH_4	Yes	

EMISSION CATEGORY	GHG SOURCE, SINK OR RESERVOIR	GHG	INCLUDED?	JUSTIFICATION
		N_2O	Yes	Release possible depending on processing method. May be demonstrated as negligible and excluded
	Embodied emissions – equipment manufacture, construction, demolition	CO_2	Yes	Primary emission from manufacture of equipment due to energy consumption
		CH_4	Yes	CH_4 release possible depending on manufacturing process and energy consumption
		N_2O	Yes	Included for completeness. May be demonstrated as negligible and excluded
	Embodied emissions - other consumables	CO_2	Yes	Primary emission from manufacture of consumables due to energy consumption
		CH_4	Yes	CH_4 release possible depending on manufacturing process and energy consumption
		N_2O	Yes	Included for completeness. May be demonstrated as negligible and excluded
Bio-oil Injection	Electricity Use	CO_2	Yes	Primary emission from electricity generation
		CH_4	Yes	Potential release during electricity generation; included for completeness
		N_2O	Yes	
	Process Emissions	CO_2	Yes	Emissions from the process (aside from energy use) are not anticipated, since bio-oil will be injected directly. Venting of the storage might be required
		CH_4	Yes	
		N_2O	Yes	

EMISSION CATEGORY	GHG SOURCE, SINK OR RESERVOIR	GHG	INCLUDED?	JUSTIFICATION
	Embodied emissions – equipment manufacture, construction, demolition	CO_2	Yes	Primary emission from manufacture of equipment due to energy consumption
		CH_4	Yes	CH_4 release possible depending on manufacturing process and energy consumption
		N_2O	Yes	Included for completeness. May be demonstrated as negligible and excluded
	Embodied emissions - other consumables	CO_2	Yes	Primary emission from manufacture of consumables due to energy consumption
		CH_4	Yes	CH_4 release possible depending on manufacturing process and energy consumption
		N_2O	Yes	Included for completeness. May be demonstrated as negligible and excluded
	Carbon in injected Bio-oil	CO_2	Yes	CO_2 stored
Transportation between Biomass Production, Conversion, & Injection	Fuel Use	CO_2	Yes	Primary emission from biomass and bio-oil transport
		CH_4	Yes	Included for completeness. May be demonstrated as negligible and excluded
		N_2O	Yes	Included for completeness. May be demonstrated as negligible and excluded
	Embodied emissions – transportation equipment manufacture, construction, demolition	CO_2	Yes	Primary emission from manufacture of equipment due to energy consumption
		CH_4	Yes	CH_4 release possible depending on manufacturing process and energy consumption

EMISSION CATEGORY	GHG SOURCE, SINK OR RESERVOIR	GHG	INCLUDED?	JUSTIFICATION
		N_2O	Yes	Included for completeness. May be demonstrated as negligible and excluded
Bio-oil storage	GHG emissions from storage formation	CO_2	Yes	Should not occur if properly designed and maintained. Included for completeness
		CH_4	Yes	Should not occur if properly designed and maintained. CH_4 may be generated by bio-oil decay. Included for completeness
		N_2O	No	Not likely to occur in subsurface environment

All GHGs must be quantified and converted to CO_2e in the GHG Statement using the 100-yr Global Warming Potential (GWP) for the GHG of interest, based on the most recent volume of the IPCC Assessment Report (currently the Sixth Assessment Report).

7.2 Baseline

The baseline scenario for bio-oil projects assumes the activities associated with the bio-oil project do not take place and any infrastructure is not built.

The counterfactual is the CO_2 stored in the biomass feedstock that would have remained durably stored in the biomass in the absence of the project. This is known as ineligible biomass, given that the CO_2 stored would have remained stored in the biomass in the absence of the CDR project and is therefore not eligible to count towards Crediting. The Biomass Feedstock Accounting Module v1.2 sets out requirements for establishing ineligible biomass as part of the Counterfactual Storage Eligibility criteria. The Biomass Feedstock Accounting Module v1.2 includes details for quantification of $CO_2e_{Counterfactual}$.

Biomass Feedstock Accounting v1.2 →

See Section 3 of the Biomass Feedstock Accounting Module for requirements.

7.3 Net CDR Calculation

7.3.1 Calculation Approach

The biomass pyrolysis process can be modeled as operating on a batch basis, consisting of a 'Production Batch', p, which typically consists of utilizing a single type of biomass feedstock, often of a single source of origin, converting the biomass to bio-oil via pyrolysis, and storing and transporting that bio-oil to an injection site. The unique characteristics of the biomass used, the bio-oil conversion process, the produced bio-oil characteristics, transportation distances, and storage site characteristics will be consistent for each Production Batch.

An 'Injection Batch', n, is a single injection activity where a quantity of bio-oil is injected into an approved underground storage site. The Injection Batch may consist of a portion of a Production Batch, a full single Production Batch, or a blend of multiple Production Batches, which is injected for durable storage.

The approach for emissions calculations here is based on Injection Batches and the specific calculation of net CO_2e removal for each Injection Batch. The following sections outline the process for calculating the net CO_2e removed for each specific Injection Batch of biomass processed and associated bio-oil injection, defined as a Removal.

The Reporting Period for a bio-oil project represents an interval of time over which removals are calculated and reported for verification. When total net CO_2e removals must be calculated for a Reporting Period, for example during submission of Claimed Removals in a GHG statement, it is calculated as the sum of removals during the Reporting Period:

$$CO_2e_{Removal,\,RP} = \sum_{1}^{n} CO_2e_{Removal,\,n}$$

(Equation 1)

Where

- $CO_2e_{Removal,\,RP}$= the total net CO_2e removal for reporting period RP, in tonnes of CO_2e
- $CO_2e_{Removal,\,n}$ = the total net CO_2e removal for Injection Batch n, occurring in reporting period RP, in tonnes of CO_2e, see Section 7.4, Equation 2

Note: Reversals occur after Credits have been issued so are not included in this equation. See Section 5.6 of the Isometric Standard for further information.

7.4 Calculation of $CO_2e_{Removal,\,n}$

Net CO_2e removal for bio-oil project can be calculated as follows. Note that the calculation is completed for a discrete batch, n, of bio-oil that is injected ('Injection Batch'). The final net CO_2e quantification must be conservatively determined, giving high confidence that at least the estimated amount of CO_2e was removed.

$$CO_2e_{Removal,\,n} = CO_2e_{Stored,\,n} - CO_2e_{Counterfactual,\,n} - CO_2e_{Emissions,\,n}$$

(Equation 2)

Where

- $CO_2e_{Stored,\,n}$= the total CO_2 removed from the atmosphere and durably stored as biogenic carbon for batch *n,* in tonnes of CO_2e, see Section 7.4.1
- $CO_2e_{Counterfactual,\,n}$= the total counterfactual CO_2 removed from the atmosphere and durably stored as biogenic carbon in the absence of the project, for batch *n,* in tonnes of CO_2e, see Section 7.4.2

- $CO_2e_{Emissions,\,n}$= the total GHG emissions for batch n, in tonnes of CO_2e, see Section 7.4.3

7.4.1 Calculation of $CO_2e_{Stored,\,n}$

CO_2e_{Stored} represents the amount of CO_2 (stored as organic carbon, C) that is injected and stored in the geologic or engineered storage formation. This is the gross amount stored for each batch injection and does not account for spillage nor reversals of storage from the storage formation.

The total amount of CO_2 contained in the injectant can be calculated as follows.

Where all bio-oil production batches are blended prior to injection:

$$CO_2e_{Stored,\,n} = \frac{C_{Bio-oil,\,n} \cdot m_{Inj,\,n}}{C_{CO_2}}$$

(Equation 3)

Where bio-oil production batches are not blended prior to injection:

$$CO_2e_{Stored,\,n} = \sum_{p=1}^{k} \left(\frac{C_{Bio-oil,\,p} \cdot m_{Inj,\,p}}{C_{CO_2}} \right)$$

(Equation 4)

Where:

- n = Injection batch

- p = Production batch
- k = number of Production batches, p, blended in the Injection batch, n
- $CO_2e_{Stored,\,n}$= the total CO_2 removed for batch n, in tonnes of CO_2e
- $C_{Bio-oil,\,n(p)}$ = the concentration as weight percent (%wt) of C in the bio-oil injectant injected for Injection Batch n OR for each production batch p included in Injection Batch n
- $m_{Inj,\,n(p)}$= the total mass of injectant emplaced via injection (tonne) for Injection Batch n OR for each production batch p included in Injection Batch n
- C_{CO_2} = the content of C in CO_2 (as a mass percent)

7.4.1.1 Measurements - $CO_2e_{Stored,\,n}$

Calculation of CO_2e_{Stored} requires two primary measurements:

- $C_{Bio-oil}$ - %wt of C in the bio-oil injectant
- m_{Inj} - total mass of injectant

7.4.1.1.1 Bio-oil Carbon Content Measurement

In order to determine $C_{Bio-oil}$, the %wt of C in the bio-oil injectant for either a blended bio-oil in Injection Batch n, or for individual Production Batches p, is determined via the analysis of samples of bio-oil injectant for total C content.

The following test method should be used where possible, and must be used for bio-oils or blends with vapor pressures higher than 3 psi[7]:

- NREL Laboratory Analysis Procedure for Determination of Carbon, Hydrogen, and Nitrogen in Bio-oils[8] should be used where possible, and must be used for bio-oils or blends with vapor pressures higher than 3 psi[7]

Where this is not possible, other acceptable test methods include:

- ASTM D5291: Standard Test Methods for Instrumental Determination of Carbon, Hydrogen, and Nitrogen in Petroleum Products and Lubricants
- ASTM D5373: Standard Test Methods for Determination of Carbon, Hydrogen and Nitrogen in Analysis Samples of Coal and Carbon in Analysis Samples of Coal and Coke
- Alternative methods or analytical equipment which determine total carbon content may be utilized if justified and documented to be equivalent to ASTM D5291.

A minimum of one sample per Injection Batch (for blended injections) or one per Production Batch (for non-blended injections) of bio-oil must be collected and analyzed. Samples shall be from a well mixed and representative aliquot of the bio-oil as injected, insuring that solids are blended in the sample and representative of the amount in the injectant.

Analysis must be completed by a qualified laboratory, as evidenced by accreditation to ISO 17025 or equivalent standards for laboratory quality management for the specific test method (ASTM D5291).

Laboratories shall complete standard quality assurance procedures on a schedule in accordance with their quality management plans and accreditation requirements to include:

- analysis of blanks
- analysis of duplicates
- instrumentation calibrations and analysis of calibration standards

This Protocol provides two alternative methods for how often C content must be measured and quantified. The first method (A) involves measuring every batch, the second method (B) involves only sampling some batches, and conservatively estimating the C content of unsampled batches.

Method A: Measure every Batch

The C content of every Injection Batch must be ascertained through direct measurement, either by:

1. Sampling the Injection Batch (which may be blended, or non-blended)
2. Sampling all constituent Production Batches which comprise the Injection Batch, and calculating the C content of the Injection Batch through a linear weighted combination of the C contents of these Production Batches, as given in *Equation 3*.
    - Note that in the case of an unblended Injection Batch, the Injection Batch originates from only a single Production Batch, by definition, and no calculation is required

For the acceptable minimum number of samples to take per sampled Batch, see Minimum number of samples per Batch below. If multiple samples are taken per Batch, the average C content of these samples must be used.

Method B: Sampling a Production Process

For a given Production Process of a feedstock, samples must be taken directly for at least 30 Production Batches, to ensure there is enough data to estimate C content for future Production Batches with appropriate statistical significance. Until this threshold is reached, Method A must be used.

Subsequently, samples must be taken at least every 10 Production Batches.

For the acceptable minimum number of samples to take per Batch, see Minimum number of samples per Batch below.

For batches which are not sampled, C content must be conservatively estimated, as follows:

$$C_{Bio-oil} = \mu_{CC} - \sigma_{\overline{CC}}$$

(Equation 5)

$$\sigma_{\overline{CC}} = \frac{\sigma_{CC}}{\sqrt{n_{samples}}}$$

(Equation 6)

where:

- $\sigma_{\overline{CC}}$ is the standard error of the mean of C content, across all eligible samples for this Production Process
- σ_{CC} is the standard deviation of C content, across all eligible samples for this Production Process
- $n_{samples}$ is the number of eligible samples for this Production Process
- μ_{CC} is the mean C content of all eligible samples for this Production Process

Eligible samples are those taken in the previous 6 months before a specific Production Batch was produced. Older samples may not be used.

Additionally, batches must be subject to random sampling, to alleviate the risk of any given batch containing a substance with a substantially different C content.

A random sampling approach must be agreed and documented in the Project Design Document, whereby Isometric will contact the Project Proponent on randomly selected days, at an agreed cadence, which must be no less frequent than once per month, on average. Once contacted, the Project Proponent must sample the C content of the subsequent batches processed.

If the Project Proponent is unable to carry this random sampling out on 3 occasions within a 6 month period, or within a 6 month period more than 3 measurements are below 3 SD from the mean, this will trigger a Project review by Isometric.

If there is a significant change to a Production Process for a feedstock, which is likely to alter the average C content of the feedstock, or if significant deviations in carbon content are detected, the feedstock should be considered as a new Production Process. This means that sampling must be restarted, with all prior samples no longer able to be used for estimating C content.

Minimum number of samples per Batch

For all measurements taken, samples must be from a well mixed and representative aliquot of the injectant. To account for the possibility of variation within a single Production Batch (for example within a large container of the injectant), either of the following approaches must be adopted:

1. A minimum of 3 samples must be taken for each measured Production Batch.
2. Justification and evidence must be provided to demonstrate that the “within batch” variation is likely to be minimal. For example, this could be justified due to the physical details of the Production Process used, or alternatively by providing data examining the “within batch” C content variation.

Process for handling C content measurement outliers

For a given Production Process, an Outlier is defined as any individual sample which lies more than 3 standard deviations, σ_{CC}, above or below the mean. To minimize the potential overall impact of outlier measurements, all C content measurement outliers must be handled via the applying the technique of "winsorization”, as follows.

For a given measurement, m, the winsorized measurement m_w is defined as follows:

- For a measurement m where $m > \mu + 3\sigma_{CC}, m_w = \mu + 3\sigma_{CC}$
- For a measurement m where $m < \mu - 3\sigma_{CC}, m_w = \mu - 3\sigma_{CC}$

The winsorized measurement, m_w, must be used for the determination of carbon content.

This winsorization process must only be applied once a minimum number of 30 measurements have been taken, to ensure statistical significance.

The Project Proponent must monitor occurrences of outliers, and investigate if significantly more than the statistically expected number occurs, as it may be indicative of a systematic issue. This may be checked at verification, at the discretion of the verifying VVB.

7.4.1.1.2 Measurement of Mass of Bio-oil Injected

The mass of injectant, m_{Inj}, is measured via determination of weight of delivered bio-oil to the injection site using a calibrated scale. The total mass injected may be determined by the difference in bio-oil delivery truck weight measured upon arrival at the injection facility and at departure, after offloading of bio-oil, either into storage or directly to injection.

Any truck scale used must have a current certification in accordance with applicable local, state, or federal regulations for legal-for-trade weights and measures. Testing and calibration of scales must utilize certified weights in accordance with local, state, or other regulations, calibration weights must meet NIST Handbook 44 specifications[9], and scale testing and calibration must be performed by a state certified entity.

The total mass injected may also be determined by other methods, such as use of a calibrated flow meter and density measurement, or use of calibrated on site weigh scales for smaller containers, where such methods are viable and justified. Note that, due to typical viscosity of bio-oil, the use of flow meters is not often viable and can result in poor data quality.

7.4.1.2 Required Records & Documentation - $CO_2e_{Stored,\,n}$

The Project Proponent must maintain the following records as evidence of CO_2 removal in injected bio-oil:

- weigh scale tickets for each delivery of bio-oil (arrival and departure weights) or other equivalent records

- analytical results for each ASTM D5291 analysis for C content of bio-oil from each batch as required
- Documentation of any spills during injection operations and estimates of quantity released

Records of all C analyses and injection masses (e.g. weigh scale tickets) must be maintained by the injection facility and provided for verification purposes for a period of five years.

7.4.1.3 Other Considerations - $CO_2e_{Stored, n}$

Although limited and of small quantity, injection processes should be monitored to ensure that any process upsets or equipment failures and resulting spills of bio-oil are monitored, documented, quantified, and accounted for in the GHG Statement of the project batch. For each batch, where a process upset results in loss of bio-oil, that amount must be deducted from the delivered amount of bio-oil based on delivery weigh tickets. Such amounts must be allocated directly to the specific injection batch of bio-oil.

7.4.2 Calculation of $CO_2e_{Counterfactual, n}$

The calculation of $CO_2e_{Counterfactual,\,n}$, is determined by the requirements of the Biomass Feedstock Accounting Module v1.2.

Biomass Feedstock Accounting v1.2 →

See Section 3 of the Biomass Feedstock Accounting Module

7.4.3 Calculation of $CO_2e_{Emissions, n}$

$CO_2e_{Emissions,\,n}$ is the total quantity of GHG emissions from operations and allocated embodied emissions for a batch n. This can be calculated as:

$$CO_2e_{Emissions,\,n} = CO_2e_{Energy,\,n} + CO_2e_{Transportation,\,n} + CO_2e_{Embodied,\,n} + CO_2e_{Misc.,\,n} + CO_2e_{Leakage,n}$$

(Equation 7)

Where

- $CO_2e_{Emissions,\,n}$= the total GHG emissions for a batch n, in tonnes of CO_2e
- $CO_2e_{Energy,\,n}$= the total GHG emissions associated with energy consumption for a batch n, in tonnes of CO_2e, see Section 7.4.3.2
- $CO_2e_{Transportation,\,n}$= the total GHG emissions associated with transportation of products for a batch n, in tonnes of CO_2e, see Section 7.4.3.3
- $CO_2e_{Embodied,\,n}$= the total embodied GHG emissions allocated to a batch n, in tonnes of CO_2e, see Section 7.4.3.4
- $CO_2e_{Misc.,\,n}$= the total miscellaneous GHG emissions for a batch n, that cannot be categorized by $CO_2e_{Energy,\,n}$, $CO_2e_{Transportation,\,n}$, or $CO_2e_{Embodied,\,n}$, in tonnes of CO_2e, see Section 7.4.3.5
- $CO_2e_{Leakage,\,n}$= the total GHG emissions associated with the project's impact on activities that fall outside of the system boundary of a project, allocated to batch n, in tonnes of CO_2e, see Section 7.4.3.6

Note: Reversals occur after Credits have been issued so are not included in this equation. See Section 5.6 of the Isometric Standard for further information. Risk of reversal information is given in Appendix 1: Risk of Reversal Questionnaire, with further information provided within the relevant storage module storage module.

7.4.3.1 Emissions allocation

Emissions that occur relating to a batch, n, must be included in the reporting of emissions associated with that batch and may not be allocated across multiple batches. Allocation

across multiple batches must be agreed with Isometric on a case by case basis.

Embodied emissions which relate to multiple batches may be allocated in line with the allocation rules set out in the Embodied Emissions Accounting Module v1.0.

When the Project Proponent is planning to cease operations within a given storage site, the monitoring emissions required for post-closure monitoring must be calculated and allocated to the remaining removals taking place at the storage site. If that is not possible, the Project Proponent should allocate those emissions to other projects and/or storage sites they conduct removal operations at, in agreement with Isometric. If for any reason emissions are not appropriately allocated, the Reversal process will be triggered in accordance with Isometric Standard, to account for any remaining monitoring emissions.

In instances where monitoring activities are shared between entities, for example if multiple companies inject bio-oil into the same storage infrastructure, the emissions associated with these activities must be allocated proportionally between the entities.

7.4.3.2 Calculation of $CO_2e_{Energy,\,n}$

GHG emissions associated with $CO_2e_{Energy,n}$ should include all emissions associated with electricity usage or fuel combustion.

Examples of electricity usage may include, but are not limited to:

- processing equipment, motors, drives and instrumentation
- facility operations
- pyrolysis system start up
- injection operations
- monitoring equipment operation, including analyzers, instrumentation, on-site laboratories specifically for monitoring activities
- sampling pumps, sampling systems, or other similar monitoring activities
- off site analytical laboratory operation and sample analysis
- electricity for building operation & management for monitoring facility buildings

Examples of fuel consumption may include, but are not limited to:

- pyrolysis system start up
- pyrolysis reactor heating
- emission control (e.g. propane for flare co-firing or pilot)
- handling equipment, such as fork trucks or loaders
- sampling system operation, such as any pumps or heating systems
- monitoring system installation, operation or closure

The Energy Use Accounting Module v1.2 provides requirements on how energy-related emissions must be calculated so that they can be subtracted in the net CO_2e removal calculation. It sets out the calculation approach to be followed for intensive facilities and non-intensive facilities and acceptable emissions factors.

Energy Use Accounting v1.2 →

Refer to Energy Use Accounting Module for the calculation guidelines.

7.4.3.3 Calculation of $CO_2e_{Transportation,\ n}$

GHG emissions associated with $CO_2e_{Transportation,n}$ should include all emissions associated with transportation of products as part of a batch n's process, including the following:

- transportation of biomass feedstock to processing site
- transportation of bio-oil from processing site to injection site
- transportation of samples for lab analysis

The Transportation Emissions Accounting Module v1.1 provides requirements on how transportation-related emissions must be calculated so that they can be subtracted in the

net CO_2e removal calculation. It sets out the calculation scope, approach to be followed, and acceptable emissions factors.

Transportation Emissions Accounting v1.1

Refer to Transportation Emissions Accounting Module for the calculation guidelines.

7.4.3.4 Calculation of $CO_2e_{Embodied, n}$

The Project Proponent must identify all equipment and consumables used in the biomass conversion and storage process, identify appropriate cradle to grave emission factors, and allocate the emissions to removals appropriately in line with the Embodied Emissions Accounting Module v1.0.

GHG emissions associated with $CO_2e_{Embodied,n}$ should include all emissions associated with the procurement and use of materials, consumables and equipment, including but not limited to:

- non-feedstock conversion process inputs or consumables:
  - catalysts
  - water
  - thermal oils used for heat transfer
  - coolants used for bio-oil quench
  - gasses such as nitrogen used for process or instrumentation purges
  - injection additives (i.e. biocide, salt, etc.)
  - gases, reagents or other materials used for operation of monitoring equipment and on-site analyzers
- equipment:
  - pyrolyzer reactor
  - feedstock conveyors, augers, feed bins, and related equipment

- bio-oil quench or condenser units, including any heat transfer equipment, such as glycol chillers and pumps
- ash and char collection systems and quench or cooling systems
- tailgas emissions control systems, such as flares or oxidizers
- preparation or mixing equipment
- pumps, piping, and related equipment
- storage tanks
- injection well materials
- all support structures, facilities, and infrastructure, including steel platforms, framing, supports, concrete footings and building structures
- monitoring wells and all associated materials (steel casing, concrete, etc.)
- on-line analyzers, measurement equipment, or other such devices
- buildings and associated equipment utilized for monitoring purposes (e.g. on-site laboratories)

Embodied Emissions Accounting v1.0

Refer to Embodied Emissions Accounting Module for the calculation guidelines.

7.4.3.5 Calculation of $CO_2e_{Misc., n}$

GHG emissions associated with $CO_2e_{Misc.,n}$ should include all emissions associated with the project that cannot be categorized by $CO_2e_{Energy,\,n}$, $CO_2e_{Transport,\,n}$, or $CO_2e_{Embodied\,n}$. The Project Proponent is responsible for identifying all sources of emissions directly or indirectly related to project activities and must report any outside of the SSR categories identified as miscellaneous emissions.

Examples include, but are not limited to:

- waste processing associated with all aspects of the project's processes
- staff travel associated with the project

- Direct emissions of tailgas as part of a pyrolysis process, $CO_2e_{Tailgas,\,n}$. Further detail is provided in the section below.

Calculation of $CO_2e_{Tailgas,\,n}$

Direct emissions from a pyrolysis process may occur when pyrolysis gasses are emitted to the atmosphere, are combusted within the process or pyrolyzer to provide thermal energy for the process, or are combusted or oxidized in an emissions control process such as a flare or thermal oxidizer.

Emissions are calculated as follows:

$$CO_2e_{Tailgas,\,p} = m_{Tailgas} \cdot C_{Tailgas,\,CH_4} \cdot GWP_{CH_4} \cdot t_p$$

(Equation 8)

Where:

- $m_{Tailgas}$ = the mass flow rate of tail gas (kg/hr)
- $C_{Tailgas,\,CH_4}$ = the concentration of CH_4 in the tail gas (wt%)
- GWP_{CH_4} = global warming potential of methane, GWP100 using IPCC Sixth Assessment Report (or most recently released IPCC report, whichever is latest)
- t_p = duration of the conversion process operation (hr) for Production Batch p

Measurement - $CO_2e_{Tailgas,\,n}$

Quantification of $CO_2e_{Tailgas}$ requires two primary measurements, the measurement of tail gas flow and the analysis of gas for CH_4 content. CO_2 content is not included as part of the $CO_2e_{Tailgas}$ emissions as the Biomass Accounting Module v1.2 requires that only biomass which would have decayed within 15 years is used. Therefore the immediate

emissions of CO_2 due to pyrolysis are not discounted against the net CO_2e removal due to the similarity of time horizons between these emissions and the counterfactual storage.

Tail gas flow rate, $m_{Tailgas}$ can be determined by various acceptable methods, including:

- use of calibrated flow meters to provide continuous volumetric or mass flow measurement. Any flow meter must be calibrated for the composition and density of tail gas[10], or use appropriate conversion factors;
- use of flow data and curves from tail gas emissions testing and pressure drop measurement (i.e. pitot tubes) in the tail gas stream. Such testing data should be produced by a qualified emissions testing company, accredited to the Stack Testing Accreditation Council for ASTM D7036, ISO 17025, or approved by state regulatory authority to perform compliance emissions testing. Testing should be completed for each type of biomass feedstock utilized unless analytical testing (ASTM D5291 ultimate analysis or proximate analysis for moisture, volatiles, fixed carbon and ash) can demonstrate that the feedstocks are similar (within 10% for each component)
- calculation of tailgas amount by a carbon material balance. Material balance calculation will require measurement of biomass weight fed to pyrolyzer and biomass C content, bio-oil product mass and bio-oil carbon content, and ash/char mass with ash/char C content for a given pyrolysis batch using calibrated weigh scales or other calibrated measurement equipment. C content analyses should be performed using ASTM D5291 or equivalent ultimate analysis procedure.

The concentration of CH_4 in the tail gas must be measured directly via one of the following methods:

- on-line analyzer measurement of CH_4 concentration, such as on-line gas chromatography, non-dispersive infrared (NDIR) detector, or similar. Analyzers must be calibrated regularly using NIST-traceable certified gas standards with concentrations of CH_4 within +/- 30% of expected average tailgas concentration;
- use of concentration data from tail gas emissions testing. Such testing data should be produced by a qualified emissions testing company, accredited to the Stack Testing Accreditation Council for ASTM D7036, ISO 17025, or approved by state

regulatory authority to perform compliance emissions testing. Emissions data should only be used when pyrolysis operating conditions are similar to the conditions under which testing was completed, and for which the biomass feedstock processed during testing is sufficiently similar (within 10% for each component, as outlined earlier).

Required Records & Documentation - $CO_2e_{Tailgas, n}$

The Project Proponent must maintain the following records as evidence supporting calculation of emissions from the biomass conversion process:

- Results of any emissions tests used to determine emission rates of CH_4 in tail gas or flow measurements of gas flow from pyrolysis process or associated emission control devices, including signed report from accredited emissions testing entity
- flow rate data from flow meters (including pitot tubes) for each period of interest (batch), including flow meter data recorded in data acquisition systems, manual operation logs, or other records indicating date, time, and flow rate, as well as meter identification number or ID

7.4.3.6 Calculation of $CO_2e_{Leakage, n}$

$CO_2e_{Leakage,\, n}$ includes emissions associated with a project's impact on activities outside the system boundary of the project. This includes instances where the Project causes an increase in GHG emissions by diverting material from other uses or incentivizing increased production activity.

It is the Project Proponent's responsibility to identify potential sources of leakage emissions. For a bio-oil project, replacement emissions must be considered as a minimum. calculations required for replacement emissions associated with market leakage are set out in the Biomass Feedstock Accounting Module v1.2.

In line with the Eligibility Criteria set out in the Biomass Feedstock Accounting Module v1.2, projects which would lead to ecological leakage associated with land use change are not eligible under this Protocol.

Biomass Feedstock Accounting v1.2

See Section 3 of the Biomass Feedstock Accounting Module

8.0 Bio-oil Storage

This Protocol provides two options for durable storage of bio-oil. The Project Proponent can choose from available options when submitting their Project for verification.

Bio-oil Storage in Permeable Reservoirs v1.1 →

Durability and monitoring requirements for storage in permeable reservoirs.

Biomass or Bio-oil Storage in Salt Caverns v1.1 →

Durability and monitoring requirements for storage in salt caverns.

9.0 Acknowledgements

Isometric would like to thank following contributors to this Protocol and relevant Modules:

- Tim Hansen (350 Solutions); Bio-oil Geological Storage Protocol and Energy Use Accounting, Transportation Emissions Accounting and Embodied Emissions Accounting Modules.
- Kevin Fingerman, Ph.D. (Cal Poly Humboldt); Biomass Feedstock Accounting Module.
- Chris Holdsworth, Ph.D. (University of Edinburgh); Bio-oil Storage in Permeable Reservoirs Module.
- Catherine Spurin, Ph.D. (Stanford University); Bio-oil Storage in Permeable Reservoirs Module.
- Wilson Ricks (Princeton University); Energy Use Accounting Module.
- Grant Faber (Carbon Based Consulting); Transportation Emissions Accounting Module.

Isometric would like to thank following reviewers of this Protocol and relevant modules:

- Sarah Saltzer, Ph.D. (Stanford University); Bio-oil Storage in Permeable Reservoirs Module.
- Grant Faber (Carbon Based Consulting); Energy Use Accounting & Embodied Emissions Accounting Modules.

10.0 Definitions and Acronyms

Activity
An activity or process or group of activities or processes that alter the condition of a Baseline and leads to Removals or Reductions.

Additionality
An evaluation of the likelihood that an intervention—for example, a CDR Project—causes a climate benefit above and beyond what would have happened in a no-intervention Baseline scenario.

Baseline
A set of data describing pre-intervention or control conditions to be used as a reference scenario for comparison.

Bio-oil
A mixture of water, organic acids, aldehydes, ketones, sugars, phenols, and other organic compounds derived from the thermal breakdown of biomass. Thermal breakdown of biomass is achieved via thermochemical processes, such as pyrolysis, which heat biomass in low- or no-oxygen environments to high temperatures (~e.g. 350-650°C). Bio-oil is often also referred to as pyrolysis oil or bio-crude.

Biomass Carbon Removal and Storage (BiCRS)
A range of processes that use biogenic material to remove carbon dioxide (CO_2) from the atmosphere and store that CO_2 underground or in long-lived products (LLNL BiCRS Roadmap, 2020).

Buyer
An entity that purchases Removals or Reductions, often with the purpose of Retiring Credits to make a Removal or Reduction claim.

Carbon Dioxide Equivalent Emissions (CO_2e)
The amount of CO_2 emissions that would cause the same integrated radiative forcing or temperature change, over a given time horizon, as an emitted amount of GHG or a mixture of GHGs. One common metric of CO_2e is the 100-year Global Warming Potential.

Carbon Dioxide Removal (CDR)
Activities that remove carbon dioxide (CO_2) from the atmosphere and store it in products or geological, terrestrial, and oceanic Reservoirs. CDR includes the enhancement of biological or geochemical sinks and direct air capture (DAC) and storage, but excludes natural CO_2 uptake not directly caused by human intervention.

Carbon Finance
Resources provided to projects that are generating, or are expected to generate, greenhouse gas (GHG) Emission Reductions or Removals.

Certification (of a Protocol)
The Isometric process which involves expert review and Public Consultation in order to arrive at an approved version of a Protocol, against which Projects will be Validated and Removals or Reductions will be Verified.

Conservative
Purposefully erring on the side of caution under conditions of Uncertainty by choosing input parameter values that will result in a lower net CO_2 Removal or GHG Reduction than if using the median input values. This is done to increase the likelihood that a given Removal or Reduction calculation is an underestimation rather than an overestimation.

Counterfactual
An assessment of what would have happened in the absence of a particular intervention – i.e., assuming the Baseline scenario.

Cradle-to-Grave
Considering impacts at each stage of a product's life cycle, from the time natural resources are extracted from the ground and processed through each subsequent stage of manufacturing, transportation, product use, and ultimately, disposal.

Credit
A publicly visible uniquely identifiable Credit Certificate Issued by a Registry that gives the owner of the Credit the right to account for one net metric tonne of Verified CO_2e Removal or Reduction. In the case of this Standard, the net tonne of CO_2e Removal or Reduction comes from a Project Validated against a Certified Protocol.

Crediting Period
The period of time over which a Project Design Document is valid, and over which Removals or Reductions may be Verified, resulting in Issued Credits.

Direct Emissions
Emissions that are produced by a specific CDR process and are directly controllable.

Double Counting
Improperly allocating the same Removal or Reduction from a Project Proponent more than once to multiple Buyers.

Durability

The amount of time carbon removed from the atmosphere by an intervention – for example, a CDR project – is expected to reside in a given Reservoir, taking into account both physical risks and socioeconomic constructs (such as contracts) to protect the Reservoir in question.

Embodied Emissions

Life cycle GHG emissions associated with production of materials, transportation, and construction or other processes for goods or buildings.

Emission Factor

An estimate of the emissions intensity per unit of an activity.

Emission Reductions

Lowering future GHG releases from a specific entity.

Enhanced Hydrocarbon Recovery (EHR)

Enhanced hydrocarbon recovery (EHR) is a tertiary hydrocarbon production technique or process where the physicochemical (physical and chemical) properties of the rock and/or the fluids are changed to enhance the recovery of hydrocarbon, typically by altering the chemical, biochemical, density, miscibility, interfacial tension (IFT)/surface tension (ST), viscosity and thermal properties to enable additional hydrocarbon production (SPE, 2023). EHR+ is the specific use of CO_2 injection for EHR where the CO_2 remains stored in the geologic formation permanently (IEA, 2015).

Feedstock

Raw material which is used for CO_2 Removal or GHG Reduction.

GHG Statement

A document submitted alongside Claimed Removals and/or Reductions that details the calculations associated with a Removal or Reduction, including the Project's emissions, Removals, Reductions and Leakages, presented together in net metric tonnes of CO_2e per Removal or Reduction.

Global Warming Potential

A measure of how much energy the emissions of 1 tonne of a GHG will absorb over a given period of time, relative to the emissions of 1 ton of CO_2.

Greenhouse Gas (GHG)

Those gaseous constituents of the atmosphere, both natural and anthropogenic (human-caused), that absorb and emit radiation at specific wavelengths within the spectrum of terrestrial radiation emitted by the Earth's surface, by the atmosphere itself, and by clouds. This property causes the greenhouse effect, whereby heat is trapped in Earth's atmosphere (CDR Primer, 2022).

International Standards Organization (ISO)
A worldwide federation (NGO) of national standards bodies from more than 160 countries, one from each member country.

Isometric Science Platform
A community resource where Project Proponents publish and visualize their early processes, Removal and Reduction data and Protocols – enabling the scientific community to share feedback and advice.

Leakage
The increase in GHG emissions outside the geographic or temporal boundary of a project that results from that project's activities.

Materiality
An acceptable difference between reported Removals/emissions or Reductions/emissions and what an auditor determines is the actual Removal/emissions or Reduction/emissions.

Project
An activity or process or group of activities or processes that alter the condition of a Baseline and leads to Removals or Reductions.

Project Design Document (PDD)
The document that clearly outlines how a Project will generate rigorously quantifiable Additional high-quality Removals or Reductions.

Project Proponent
The organization that develops and/or has overall legal ownership or control of a Removal or Reduction Project.

Protocol
A document that describes how to quantitatively assess the net amount of CO_2 removed by a process. To Isometric, a Protocol is specific to a Project Proponent's process and comprised of Modules representing the Carbon Fluxes involved in the CDR process. A Protocol measures the full carbon impact of a process against the Baseline of it not occurring.

Registry
A database that holds information on Verified Removals and Reductions based on Protocols. Registries Issue Credits, and track their ownership and Retirement.

Removal

The term used to represent the CO_2 taken out of the atmosphere as a result of a CDR process.

Replacement Emissions
Any emissions that occur to compensate for biomass that was previously serving another purpose and is now being used for carbon removal or GHG reduction. For example, if agricultural waste was previously left on a field to decompose - fertilizer production to replace those nutrients need to be accounted for.

Reservoir
A location where carbon is stored. This can be via physical barriers (such as geological formations) or through partitioning based on chemical or biological processes (such as mineralization or photosynthesis).

Reversal
The escape of CO_2 to the atmosphere after it has been stored, and after a Credit has been Issued. A Reversal is classified as avoidable if a Project Proponent has influence or control over it and it likely could have been averted through application of reasonable risk mitigation measures. Any other Reversals will be classified as unavoidable.

Sensitivity Analysis
An analysis of how much different components in a Model contribute to the overall Uncertainty.

Sink
Any process, activity, or mechanism that removes a greenhouse gas, a precursor to a greenhouse gas, or an aerosol from the atmosphere.

Standards (scientific)
Standard physical constants as well as standard values set forth by bodies such as the National Institute of Standards and Technology (NIST) or others.

Storage
Describes the addition of carbon dioxide removed from the atmosphere to a reservoir, which serves as its ultimate destination. This is also referred to as “sequestration”.

Uncertainty
A lack of knowledge of the exact amount of CO_2 removed by a particular process, Uncertainty may be quantified using probability distributions, confidence intervals, or variance estimates.

Validation

A systematic and independent process for evaluating the reasonableness of the assumptions, limitations and methods that support a Project and assessing whether the Project conforms to the criteria set forth in the Isometric Standard and the Protocol by which the Project is governed. Validation must be completed by an Isometric approved third-party (VVB).

Validation and Verification Bodies (VVBs)
Third-party auditing organizations that are experts in their sector and used to determine if a project conforms to the rules, regulations, and standards set out by a governing body. A VVB must be approved by Isometric prior to conducting validation and verification.

Verification
A process for evaluating and confirming the net Removals and Reductions for a Project, using data and information collected from the Project and assessing conformity with the criteria set forth in the Isometric Standard and the Protocol by which it is governed. Verification must be completed by an Isometric approved third-party (VVB).

11.0 Appendix 1: Monitoring Plan Requirements

This appendix details how the Project Proponent must monitor, document and report all metrics identified within this Protocol. Following this guidance will ensure the Project Proponent measures and confirms carbon dioxide removal and long-term storage compliance, and will enable quantification of the emissions removal resulting from the Project activity during the Project Crediting Period, prior to each Verification.

This methodology utilizes a comprehensive monitoring and documentation framework that captures the GHG impact in each stage of a Project. Monitoring and detailed accounting practices must be conducted throughout to ensure the continuous integrity of the carbon removals and crediting.

The Project Proponent must develop and apply a monitoring plan according to ISO 14064-2 principles of transparency and accuracy that allows the quantification and proof of GHG emissions removals.

11.1 Modular requirements

The Modules associated with this Protocol have their own set of required parameters that need to be monitored. Please refer to the following Sections of the Modules to see a complete list of all requirements:

- Biomass Feedstock Accounting Module v1.2
- Biomass or Bio-oil Storage in Salt Caverns Module v1.1
- Bio-oil Storage in Permeable Reservoirs Module v1.1

11.2 Net CDR Calculation Requirements

These parameters must be monitored for the purpose of Carbon Emissions Calculation and Embodied Carbon Emissions Calculation.

PARAMETER	PARAMETER DESCRIPTION	REQUIRED	EQUATION	PARAMETER TYPE
$C_{Bio-oil,\,n}$	%wt of C in the bio-oil injectant	Always	Eq. 4 (Bio-oil Geological Storage)	Measured
$m_{Inj,\,n}$	Total mass of bio-oil injectant	Always	Eq. 4 (Bio-oil Geological Storage)	Measured
$C_{Bio-oil,\,p}$	%wt of C in the bio-oil injectant	Under certain conditions: • if bio-oil production batches are not blended prior to injection	Eq. 4 (Bio-oil Geological Storage)	Measured
$m_{Inj,\,p}$	Total mass of bio-oil injectant	Under certain conditions: • if bio-oil production batches are not blended prior to injection	Eq. 4 (Bio-oil Geological Storage)	Measured

PARAMETER	PARAMETER DESCRIPTION	REQUIRED	EQUATION	PARAMETER TYPE
$m_{Tailgas}$	Mass flow rate of tail gas from biomass pyrolysis	Always	Eq. 8 (Bio-oil Geological Storage)	Measured or calculated
$C_{Tailgas,\,CO_2}$	Concentration of CO_2 in tailgas	Always	Eq. 8 (Bio-oil Geological Storage)	Measured

PARAMETER	PARAMETER DESCRIPTION	REQUIRED	EQUATION	PARAMETER TYPE
$C_{Tailgas,\,CH_4}$	Concentration of CH_4 in tailgas	Always	Eq. 8 (Bio-oil Geological Storage)	Measured
t_p	Biomass conversion batch operation time	Always	Eq. 8 (Bio-oil Geological Storage)	Measured

PARAMETER	PARAMETER DESCRIPTION	REQUIRED	EQUATION	PARAMETER TYPE
$m_{fuel,\,conversion}$	Mass of fuel used in biomass conversion	Always	Eq. 6 (Energy Use Accounting Module v1.2)	Measured
$EF_{Fuel,Conversion}$	Fuel emission factor for biomass conversion	Always	Eq. 6 (Energy Use Accounting Module v1.2)	Estimated
$kwh_{Conversion}$	Electricity usage for biomass conversion	Always	Eq. 2 (Energy Use Accounting Module v1.2)	Measured
$EF_{Elect,Conversion}$	Electricity emission factor	Always	Eq. 2 (Energy Use Accounting Module)	Estimated
$F_{Conversion}$	Quantity of fuel used in transport of biomass to	Under certain conditions	Eq. 2 (Transportation Emissions	Measured or estimated

PARAMETER	PARAMETER DESCRIPTION	REQUIRED	EQUATION	PARAMETER TYPE
	conversion site		Accounting Module v1.1)	
$F_{Injection}$	Quantity of fuel used in transport of bio-oil to injection site	Under certain conditions	Eq. 2 (Transportation Emissions Accounting Module v1.1)	Measured or estimated
$EF_{Fuel,Transportation}$	Fuel emission factor for transportation	Under certain conditions	Eq. 2 (Transportation Emissions Accounting Module v1.1)	Estimated
$D_{Conversion}$	Biomass transportation distance traveled - feedstock supplier to conversion site	Under certain conditions	Eq. 2 (Transportation Emissions Accounting Module v1.1)	Measured or estimated

PARAMETER	PARAMETER DESCRIPTION	REQUIRED	EQUATION	PARAMETER TYPE
$W_{Conversion}$	Mass of biomass transported from supplier to conversion site	Under certain conditions	Eq. 3 (Transportation Emissions Accounting Module v1.1)	Measured
$D_{Injection}$	Bio-oil transportation distance traveled - conversion site to injection site	Under certain conditions	Eq. 3 (Transportation Emissions Accounting Module v1.1)	Measured or estimated
$W_{Injection}$	Mass of bio-oil transported from conversion site to injection site	Under certain conditions	Eq. 3 (Transportation Emissions Accounting Module v1.1)	Measured
$EF_{Transportation,j}$	The weight- and distance-based emission factor for transportation	Under certain conditions	Eq. 3 (Transportation Emissions Accounting Module v1.1)	Estimated

PARAMETER	PARAMETER DESCRIPTION	REQUIRED	EQUATION	PARAMETER TYPE
$m_{fuel,\,Injection}$	Mass of fuel used in bio-oil injection	Always	Eq. 6 (Energy Use Accounting Module v1.2)	Measured
$EF_{Fuel,Injection}$	Fuel emission factor for bio-oil injection	Always	Eq. 6. (Energy Use Accounting Module v1.2)	Estimated
$kwh_{Injection}$	Electricity usage for bio-oil injection	Always	Eq. 2 (Energy Use Accounting Module v1.2)	Measured

PARAMETER	PARAMETER DESCRIPTION	REQUIRED	EQUATION	PARAMETER TYPE
$EF_{Elect,Injection}$	Electricity emission factor	Always	Eq. 2 (Energy Use Accounting Module v1.2)	Estimated
Product Stage Emissions	Includes raw material sourcing, transport to facility and manufacturing	Always		Measured
Construction Stage Emissions	Includes transport to site and installation at site	Always		Measured

PARAMETER	PARAMETER DESCRIPTION	REQUIRED	EQUATION	PARAMETER TYPE
End of Life Stage Emissions	Includes demolition of building, transport to end of life, waste processing and final disposal or scenarios for these life cycle stages	Always		Measured
Storage and Monitoring Emissions	The total quantity of GHG emissions associated with storage monitoring operations allocated to a removal	Always		Measured

12.0 Appendix 2: Risk of Reversal Questionnaire

This risk assessment identifies the pathway specific risk factors relevant to a carbon removal project. The relevant risk factors identified as part of a risk assessment are included in the monitoring plan requirements for the project, with details included in the Project Design Document. Project specific risk factors inform the required duration of monitoring along with the monitoring requirements set out in the Protocol and the requirements set out in the Monitoring Section of the Isometric Standard.

The risk score, as determined by the Risk of Reversal Questionnaire, will determine a project's buffer pool contribution. Projects must re-assess their reversal risk at the renewal of each crediting period, or if monitoring identifies a reversal-related risk, or if an actual reversal event takes place. In any event, projects should reassess their reversal risk at a minimum every 5 years.

The Risk of Reversal Questionnaire questions that pertain to this protocol, drawn from the programme-level Risk of Reversal Questionnaire defined in Appendix B: Risk Reversal Questionnaire of the Isometric Standard, include the following:

# IN ISOMETRIC STANDARD QUESTIONNAIRE	QUESTION	IF ANSWERED "YES"	IF ANSWERED "NO"
1	Is a reversal directly observable with a physical or chemical measurement as opposed to a modeled result?	Proceed to questions 2-9	Proceed to questions 8-9
2	Is the carbon being stored in an impermeable geologic system? (e.g., salt cavern)	Proceed to questions 8-9	Add 1 to Risk Score and proceed to questions 3-9
3	Is the carbon being stored organic?	Add 1 to Risk Score	

# IN ISOMETRIC STANDARD QUESTIONNAIRE	QUESTION	IF ANSWERED "YES"	IF ANSWERED "NO"
4	Are conditions for methane production present (anaerobic conditions, lignin content)?	Add 1 to Risk Score	
5	Does this approach have a material risk of reversal due to natural disasters including, but not limited to, floods, storms, earthquakes, fires, etc.?	Add 1 to Risk Score	
6	Does this approach have a material risk of reversal due to human-induced events from outside actors, such as change in farming practices, change in ownership and management of project sites, or similar?	Add up to 2 to Risk Score	
7	**Applicable only for subsurface storage**: Is the carbon being stored with trapping mechanisms preventing reversals? (e.g., multiple confining layers, CO_2 dissolves or solidifies)	Minus 1 to Risk Score (unless 0)	
8	Is there 10+ years of monitoring and/or lab data demonstrating low project risk?	Minus up to 2 to Risk Score	
9	Does this pathway have a documented history of reversals?	Add 2 to Risk Score	
10	Is there one or more project-specific factors that merit a high risk level?	Add up to 2 to Risk Score	

Risk Score Categories

- 0: Very Low Risk Level (2% buffer)
- 1-2: Low Risk Level (5% buffer)
- 3-4: Medium Risk Level (7% buffer)
- 5+: High Risk Level (10-20% buffer)

Project specific risk factors will depend on the form of carbon being stored (i.e., organic vs. inorganic), the method of storage (e.g., mineralization, encapsulation), the location of carbon storage (e.g., subsurface, ocean), and the proximity of that carbon to potential agents of reversal.

For projects with carbon storage as organic carbon, the presence the following risk factors must be reflected in the risk score corresponding to question 10:

- High-potential oxidants
- Temperatures in excess of thermal stability
- Fires
- Microbial degradation

For projects with any form of subsurface carbon storage, the presence of the following risk factors must be reflected in the risk score corresponding to question 10:

- Seismicity
- Subsurface migration

13.0 Relevant Works

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1. https://charmindustrial.com/faq ↩
2. Sustainable sourcing may be demonstrated via certification or demonstrated compliance with programs such as the Forest Stewardship Council (https://fsc.org/en), High Conservation Value Network (https://www.hcvnetwork.org/), Sustainable Biomass Program (https://sbp-cert.org/), Roundtable for Sustainable Biomass (https://rsb.org/the-rsb-standard/about-the-rsb-standard/), European Union Renewable Energy Directive (RED II), Sustainable Forestry Initiative, or other similar biomass sustainability protocols, standards, and compliance methods. ↩
3. Note that other well classes may be utilized, such as Class II wells, if site specific UIC well permits identify bio-oil as an acceptable injectant. However, Class II wells may not be utilized if the wells are also used for enhanced hydrocarbon recovery (EHR or EHR+) activities. ↩
4. For Class V wells, the well must be permitted and not 'authorized by rule', and must consider the specific emplacement and durable storage of bio-oil in the geologic reservoir. As of writing, the utilization of Class V wells should be limited to wells operating under the Other / Experimental category of Class V wells or other appropriate well type as approved by the UIC permitting authority. ↩
5. ISO 14064-3: 2019, Section 5.1.7 ↩ ↩ [2]
6. https://carboncreditquality.org/methodology.html ↩
7. https://www.nrel.gov/docs/fy22osti/80967.pdf ↩ ↩ [2]
8. https://www.nrel.gov/docs/fy22osti/80967.pdf ↩
9. https://www.nist.gov/pml/owm/publications/nist-handbooks/handbook-44-current-edition ↩
10. Flow meters must be calibrated to national traceable standards by an ISO 17025 accredited metrology laboratory. Flow meters may include critical nozzle flow meters (i.e. ISO 9300:2022 compliant meters), coriolis mass flow meters, and other applicable meters for mixed gas flows, as long as properly calibrated and maintained. ↩

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v1.2

PENDING CERTIFICATION

Key requirement changes to GHG Accounting, including the centralisation of requirements into the GHG Accounting Module v1.0.

v1.1

CERTIFIED

New version release in accordance with that of the Biomass or Bio-oil Storage in Salt Caverns Module (v1.1) and the Bio-oil Storage in Permeable Reservoirs Module (v1.1).

1.1.2

07 Aug 2025

- Addition of appendix to clarify risk of reversal and buffer pool size

1.1.1

04 Feb 2025

- The Energy Use Accounting Module was updated to version 1.2 and the Transportation Emissions Accounting Module was updated to version 1.1

1.1.0

13 Sep 2024

v1.0

CERTIFIED

Initial Release.

1.0.7

27 Jun 2024

- Fix reference to correct version of the Energy Use Accounting Module (v1.1)
- Text formatting updates

1.0.6

22 May 2024

- Text formatting updates

1.0.5

15 May 2024

- Text formatting updates

1.0.4

01 May 2024

- Text formatting updates

1.0.3

26 Apr 2024

- GHG terminology updated
- Guidance added to the Applicability Section, see Section 4 for further details
- Guidance added to the Calculation Approach Section, see Section 7.3.1 for further details
- Guidance added to Calculation of CO2eEmissions, n Section, see Section 7.3.5 for further details
- Guidance added through creation of the Emissions Allocation Section, see Section 7.3.5.1 for further details
- Structural changes to content, Section 7.3.5.1 created

1.0.2

11 Apr 2024

- Guidance added to the Data Sharing Section, see Section 6.6 for further details
- Guidance added to the System Boundaries & GHG Emissions Scope Section, see Section 7.1 for further details
- Guidance added to the Baseline Section, see Section 7.2 for further details
- Guidance added to the Calculation of CO2eRemoval, n Section, see Section 7.4 for further details
- GHG terminology updated
- Parameters renamed in Table 1
- Parameter renamed in Equation 2
- Structural changes to content, Section 7.2.1 merged with Section 7.2 and Sections 7.4.1.1.2, 7.4.3.4, and 7.4.3.6 created
- Non-material changes to calculation structure, in Equation 7 the parameter was “CO2eLeakage, n” added
- Text formatting updates

1.0.1

05 Mar 2024

- GHG terminology updated
- Text formatting updates

1.0.0

17 Jan 2024

Changes as a result of Public Consultation:

- Requirement added to the Environmental and Social Safeguarding Section, Project Proponents must demonstrate a risk assessment and mitigation plan for safeguarding working conditions, see Section 5 for further details
- Requirement added to the Data Sharing Section, GHG statements will be publicly shown on the Isometric Registry, see Section 6.6 for further details
- Requirement changed in the System Boundary & GHG Emissions Scope Section and Calculation of CO2eRemoval, n Section, scope expanded to allow for injection of bio-oil mixed with biochar, see Sections 7.1 and 7.4.1.1 for further details
- Guidance added to the Uncertainty Section, see Section 6.5 for further details
- Guidance added to the System Boundary & GHG Emissions Scope Section, see Section 7.1 for further details
- Guidance added to the Calculation of CO2eRemoval, n Section, see Section 7.4 for further details
- Guidance added through creation of Appendix 1: Monitoring Plan Requirements, see Section 11 for further details
- Text formatting updates

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